

# APT3065 MID-TERM PROJECT

## RepairTrack: Week 13 Final Project Completion Report

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### 1. Project Overview

RepairTrack is a secure, lightweight, mobile-friendly web-based device repair tracking system designed for small ICT repair businesses. The final Week 13 prototype centralizes device intake, technician assignment, repair workflow updates, diagnostic notes, customer status lookup, reporting, and audit logging.

The system addresses the gap between complex enterprise helpdesk tools and the practical needs of small repair shops that often rely on notebooks, WhatsApp messages, or spreadsheets to track customer devices.

### 2. Completed Week 1-13 Deliverables

| Week(s) | Completed Workload                                                                            | Evidence in Package                                            |
|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1-2     | Requirements gathering, stakeholder analysis, literature review, and initial scope definition | docs/Concept Paper and Requirements documentation              |
| 2-3     | System design: architecture, UML diagrams, ERD, normalized database design, and wireframes    | docs/System Design Document                                    |
| 4-5     | Sprint 1: authentication, role-based access, and device intake module                         | public/, src/, data/, tests/                                   |
| 5-6     | Sprint 2: technician assignment, workflow status updates, notes, and audit trail              | App prototype and Week 6 report                                |
| 7-8     | Sprint 3: customer lookup, QR/reference receipt area, and service history retrieval           | public/index.php customer lookup and ticket pages              |
| 9-10    | Sprint 4: reports, CSV export, security hardening, and privacy controls                       | reports page, CSV export, CSRF, hashing, escaping, role checks |
| 11-12   | Testing, debugging, smoke                                                                     | tests/smoke_test.php and docs                                  |

|    |                                                         |                                   |
|----|---------------------------------------------------------|-----------------------------------|
|    | tests, documentation, and GitHub preparation            | guides                            |
| 13 | Final packaging, final report, and submission checklist | README, final report, project ZIP |

### 3. Final Functional Features

- Secure staff login for administrator and technician users.
- Role-based access control so administrators and technicians have different permissions.
- Device intake form for capturing customer, device, issue, accessory, priority, and due-date details.
- Automatic repair ticket reference generation using an RT- prefix.
- Ticket search, filtering, technician assignment, and status workflow management.
- Diagnostic and repair notes linked to users and timestamps.
- Customer public lookup using a repair reference number.
- QR/reference receipt display area for customer status checking.
- Administrator reports showing ticket totals, open jobs, ready-for-collection jobs, overdue jobs, and technician workload.
- CSV export for repair reports.
- Audit trail for login, ticket creation, ticket updates, and notes.

### 4. Security and Privacy Controls

- Passwords are stored using PHP password\_hash rather than plain text.
- Sessions are regenerated during login to reduce session-fixation risk.
- POST forms use CSRF token verification.
- Output is escaped before display to reduce cross-site scripting risk.
- Technicians can only view tickets assigned to them unless they have administrator privileges.
- Customer lookup displays limited repair status only and hides internal repair notes.
- Real customer data should not be uploaded to GitHub; the package includes demo data only.
- Audit logs support accountability by recording important user actions.

### 5. Technology Stack and Architecture

| Layer           | Technology / Design Choice          | Purpose                                                                 |
|-----------------|-------------------------------------|-------------------------------------------------------------------------|
| Presentation    | HTML, CSS, mobile-responsive layout | User interface for staff and customer lookup                            |
| Application     | PHP 8+                              | Routing, forms, validation, authentication, reports, and business logic |
| Data            | JSON demo storage                   | Simple classroom execution without database installation                |
| Target Database | MySQL 3NF schema                    | Future migration to                                                     |

|                    |                              |                                                       |
|--------------------|------------------------------|-------------------------------------------------------|
|                    |                              | production-ready relational storage                   |
| Version Control    | Git and GitHub               | Repository hosting, branches, commits, and submission |
| Project Management | GitHub Projects Kanban board | Sprint planning and issue tracking                    |

The design follows an MVC-style separation: public pages provide the view layer, src files contain controllers/services/security logic, and data/database files represent the model layer. This is appropriate for a small web application because it supports maintainability while avoiding unnecessary microservice complexity.

## 6. How to Run the Application

1. Install PHP 8 or later.
2. Extract the project ZIP file.
3. Open a terminal in the project folder.
4. Run `php scripts/init_demo.php` to reset and seed demo data.
5. Run `php -S localhost:8000 -t public` to start the development server.
6. Open `http://localhost:8000` in a browser.
7. Login as `admin/admin123` or `tech/tech123`.
8. Run `php tests/smoke_test.php` to verify core functions.

## 7. Testing Summary

| Test Area                              | Result |
|----------------------------------------|--------|
| PHP syntax check                       | Passed |
| Login and demo user verification       | Passed |
| Ticket reference generation            | Passed |
| Ticket creation and retrieval          | Passed |
| Ticket status update                   | Passed |
| Audit event recording                  | Passed |
| Login and lookup page HTTP smoke check | Passed |

## 8. Final Submission Checklist

- Project source code included.
- README.md included with run instructions.
- Run and GitHub submission guide included.
- Final system design document included in docs folder.
- Concept paper included in docs folder.
- MySQL target schema included.
- Smoke tests included.
- GitHub issue and pull request templates included.

- Demo data included; real customer data excluded.
- Student ID placeholder must be replaced before final upload.

## 9. Conclusion

The Week 13 package completes the planned RepairTrack prototype and supporting documentation. The system is ready for local demonstration, GitHub upload, and academic submission after the student ID and GitHub URL placeholders are updated.